Demonstrating the physics of metamaterials with Acoustic waves and Soda Cans

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Acoustic metamaterials allow for accurate control of acoustic waves due to carefully designed subwavelength structures. In the experiment, soda cans are used as subwavelength Helmholtz resonators that serve as acoustic meta-atoms. Frequency sweep experiments in the range from 200 Hz to 400 Hz result in the emergence of specific peaks in amplitude along with phase shifts crossing 90° at the resonance frequency (312.0 Hz for an empty 500 mL can and 337.3 Hz when filled with 50 mL of water). Systematic changes in the volume of internal air cavities lead to the theoretical inverse square-root scaling law between resonance frequency and volume of internal air cavities ($f_0 \propto 1/\sqrt{V_0}$) resulting in the slope value of $6.62 \pm 0.03\,\mathrm{Hz}\cdot\mathrm{m}^{3/2}$ ($R^2 = 0.994$). In addition, two 4×3 soda-can arrays with 17 cm separation were placed on the rotary table to form a directional acoustic meta-antenna. Angular dependence of the polar radiation profile at 498 Hz and 612 Hz frequencies was analyzed using the interference of three sources model.

I. OBJECTIVE

1. To study physics of a soda can as a Helmholtz resonator by measuring the amplitude and phase response of the soda-can resonator as a function of the driving frequency.

2. To investigate the resonance frequency of the soda can for the different internal volumes.

3. To construct and characterize an acoustic antenna capable of redirecting sound preferentially in some particular direction.

II. APPARATUS

Soda cans of internal volume 500 ml and 200 ml, Water and two beakers of volume 500 ml and 250 ml, Microphone, K5 behringer loudspeakers and microphone amplifier, Function generator, Oscilloscope and PicoScope, Rotating table, Retort stands and clamps, ruler and BNC cables.

III. INTRODUCTION

Metamaterials are artificially designed materials whose properties are different from the chemical composition of its base substances, but from their deliberately designed internal structure. These properties are often rare or absent in naturally occurring materials. Their shape, geometry, size, orientation, and arrangement give them their properties of manipulating electromagnetic, acoustic, or seismic waves: by blocking, absorbing, enhancing, or bending waves. There are multiple examples of metamaterials like invisibility cloaks, Superlenses, Antennas, metasurface holograms, and media with negative refractive indices. Meta-atoms are subwavelength resonators and the arrays of these meta-atoms collectively create the overall desired effect on freely propagating waves.
Soda Can manipulate and control sound waves using their geometry, spatial arrangement, and internal resonance mechanics. In this experiment, Acoustic metamaterials studied using Soda Can which mimic the Helmholtz resonator behavior. At acoustic frequencies, their resonance properties (amplitude and phase) were studied and also these properties were tuned by reducing the volume of air inside the Can. In the last part of the experiment, Soda Cans were arranged in arrays so that they can function as an acoustic metamaterial together. With the ability to control their scattering amplitude and phase, sound waves can be controlled. Based on this idea, we create an acoustic antenna and study its asymmetric polar distribution response.[1]

IV. THEORY

A. Part 1: Soda Can as Helmholtz Resonator

Helmholtz resonance refers to the phenomenon of air resonance in a cavity which occurs when air is forced in and out of a cavity (the resonance chamber), causing the air inside to vibrate at a specific natural frequency. The underlying principle involves the vibration of the air mass in the neck of the resonator, acting analogously to a mass on a spring. When external forces, such as airflow, disturb this air mass, it oscillates and causes the air within the cavity to resonate. This phenomenon is characterized by its sharp and high-amplitude resonance curve, making it distinct from other types of acoustic resonance.[2].
Soda cans behave as Helmholtz resonators whose resonances arise from acoustic oscillations of an effective mass of air near the can's opening due to changing pressure difference near it, as shown in Fig. 1

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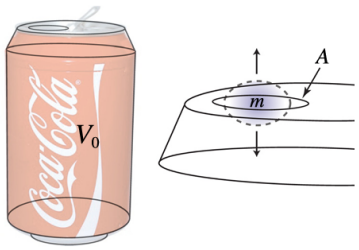


FIG. 1: A soda can is a Helmholtz resonator as the effective mass m of air near its opening oscillates due to the difference in air pressure.

The position X(t) of the effective mass of air may be described by an equation of a damped harmonic oscillator driven by a pressure difference $\Delta P e^{i\omega t}$

$$\ddot{X} + \gamma_D \dot{X} + \omega_0^2 X = g\Delta P e^{i\omega t}, \tag{1}$$

where g is a coupling constant between the pressure wave and the air, ω_0 is the resonance frequency, and γ_D is the damping constant. For the damped harmonic oscillator equation, the resonance frequency ω_0 and its solution complex amplitude X(ω) is given by

$$\omega_0^2 = \frac{A^2 \gamma P_0}{m V_0} \tag{2}$$

$$X(\omega) = \frac{g\Delta P}{\omega_0^2 - \omega^2 + i\omega\gamma_D} \tag{3}$$

where A is the area of the cavity's opening, P_0 is the ambient pressure, m is the effective mass of the air, V_0 is the resonator's internal volume, and γ is the adiabatic constant (1.4 for air).

$$|X(\omega)| = \frac{C}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(\gamma_D\,\omega\right)^2}}, \qquad C = g\,\Delta P, \tag{4}$$

The above Eq. 4 is used to fit resonance data (single harmonic oscillator). The phase shift ϕ between the motion of the air relative to the incident driving wave is

$$\phi = \tan^{-1}\left(\frac{-\omega\gamma_D}{w_0^2 - w^2}\right) \tag{5}$$

From the equation 2, it can be seen that $\omega_0 \propto \frac{1}{V_0^{1/2}}$ which means that filled Soda can has higher resonance frequency than the empty one. Experimentally, this is verified in this work.

B. Methodology

In this part of the experiment, we tried two approaches: (i) using Picoscope and (ii) using oscilloscope with function generator. A single can is placed an arbitrary distance (>15cm) away from K5 behringer loudspeaker with a microphone held over the can's neck (as shown in the figure 2). With Picoscope, resonance and phase response of empty can was studied by varying frequency from 200 Hz to 400 Hz in 5 Hz steps. Since the obtained results include some errors or undesired plots (reason discussed below), later oscilloscope and function generator was used to take data manually.

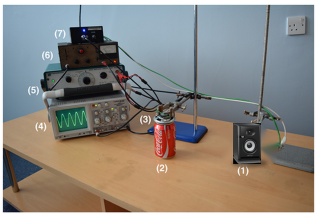


FIG. 2: Experimental apparatus: (1) source (loudspeaker); (2) can under investigation; (3) receiver (microphone); (4) oscilloscope; (5) frequency generator; (6) microphone amplifier.

C. Part 2: Multiple cans or array of cans behaves as antenna

The directional nature of a speaker's emitted sound wave can be reshaped using just two Helmholtz resonators which are out of phase. However, using only one soda can is small relative to the wavelength and interacts weakly with the sound. To counter this problem, the meta-atoms consist of 12 soda cans arranged in a 4 × 3 arrangement which helps in increasing the magnitude of the array's interaction with the sound wave.
Model assumptions: The array is considered to be a single Helmholtz resonator whose phase is constant throughout the array. When such an array is excited with sound waves, the sound that the array emits will be characterized as a spherical wave. The loudspeaker (6.4 cm in diameter) is also much smaller than the wavelength, so it can be approximated by a monopole and so, described by a spherical wave.
Based on these assumptions, the pressure variation $\Delta P(\mathbf{r})$ of a source located at $\mathbf{r}_0$, evaluated at a far-field position $\mathbf{r}$, is

$$\Delta P(\mathbf{r}) = \mathrm{Re}\left\{A\, e^{i\phi}\, \frac{e^{ik|\mathbf{r}-\mathbf{r}_0|}}{|\mathbf{r}-\mathbf{r}_0|}\right\}, \tag{6}$$

where $k = 2\pi/\lambda$ is the wavenumber, A is the amplitude of the harmonic wave, and ϕ is its phase shift relative to the source driving wave.
The antenna consists of three sources (Fig. 3): the loudspeaker at the origin ($\mathbf{r} = 0$), and two can arrays that scatter the loudspeaker's sound, placed along the z-axis at $z = \pm\delta z$. The total pressure at a position $\mathbf{r}$ far from

the loudspeaker is

$$\Delta P(\mathbf{r}) = \mathrm{Re}\left\{ A_0 \frac{e^{ik|\mathbf{r}|}}{|\mathbf{r}|} + A_1 e^{i\phi_1} \frac{e^{ik|\mathbf{r}-\delta\mathbf{z}|}}{|\mathbf{r}-\delta\mathbf{z}|} + A_2 e^{i\phi_2} \frac{e^{ik|\mathbf{r}+\delta\mathbf{z}|}}{|\mathbf{r}+\delta\mathbf{z}|} \right\}, \tag{7}$$

where $\phi_{1,2}$ are the phases of the can arrays relative to the loudspeaker phase, labelled 1 for the positive side ($z > 0$) and 2 for the negative side ($z < 0$).

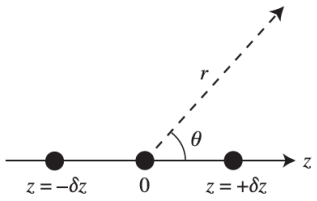


FIG. 3: Geometry used to describe the three sources: the loudspeaker at the origin and the two can arrays at $z = \pm\delta z$.

In the far-field limit ($\delta z \ll r$),

$$|\mathbf{r} \pm \delta\mathbf{z}| = \sqrt{(z \pm \delta z)^2 + y^2 + x^2} \approx r \pm (z/r)\,\delta z.$$

Using spherical coordinates with $z = r\cos\theta$, this becomes $|\mathbf{r} \pm \delta\mathbf{z}| \approx r \pm \delta z \cos\theta$. Substituting into Eq. (7) (with the same approximation applied in the phase and the denominators) gives

$$\begin{aligned} \Delta P(r,\theta) = \mathrm{Re}\Big\{ A_0 \frac{e^{ikr}}{r} \Big[1 + \frac{A_1}{A_0} e^{i\phi_1} \frac{e^{-ik\delta z\cos\theta}}{1-(\delta z/r)\cos\theta} \\ + \frac{A_2}{A_0} e^{i\phi_2} \frac{e^{ik\delta z\cos\theta}}{1+(\delta z/r)\cos\theta} \Big] \Big\} \\ \equiv \mathrm{Re}\Big\{ A_0 \frac{e^{ikr}}{r} f(\theta) \Big\}. \end{aligned} \tag{8}$$

This shows that the two arrays modify the loudspeaker's signal in an angular-dependent manner, through the directional function $f(\theta)$.

Assuming $A_1 \approx A_2$ and $\delta z \ll r$ (so the denominators in Eq. (8) are ≈ 1), the directional function is approximated by

$$f(\theta) \approx 1 + \frac{A_1}{A_0} \left[e^{i(\phi_1 - k\,\delta z\cos\theta)} + e^{i(\phi_2 + k\,\delta z\cos\theta)} \right]. \tag{9}$$

Thus $f(\theta)$ can be modified by changing the distance $2\delta z$ between the two can arrays, or by changing their resonance phases ϕ_1 and ϕ_2. With a proper choice of ϕ_1, ϕ_2 and δz, the sum of the waves propagating along one direction ($\theta = 0$) can be made greater than the sum of the waves propagating in the opposite direction ($\theta = \pi$), creating a directional antenna.Because soda cans, as single harmonic oscillators, can only be de-phased by a maximum of π, only the upper-left region of the (ϕ_1, ϕ_2) maps is physically accessible.

$$\begin{aligned} |f(\theta)|^2 = 1 + 4a\cos b\,\cos(c-x) + 4a^2\cos^2(c-x)\,, \\ x = k\,\delta z\cos\theta. \end{aligned} \tag{10}$$

where $a = A_1/A_0$, $b = (\phi_1 + \phi_2)/2$ and $c = (\phi_1 - \phi_2)/2$. This Eq. 10 is used to fit polar plots.

D. Methodology

The acoustic antenna was constructed from two 4×3 soda-can arrays mounted on a rotating table, with a 6.4 cm loudspeaker above the center and a microphone positioned more than 40 cm away. Asymmetric directionality can be obtained by setting the array phases ϕ_1 and ϕ_2 according to the phase maps.
In paper [1], the two arrays were detuned by a phase shift of π, so that one array interferes constructively with the loudspeaker's wave while the other interferes destructively.
In our experiment, one set of cans was filled with 20 ml of water and the other with 28.11 ml. The required volumes were found by first filling two cans with the same amount of water(25 ml each) and then transferring water from one can to the other so that we could get phase difference of $\pi/2$. But we were not able to get phase difference $\pi/2$. The speaker was driven at two frequencies 498 Hz and 612 Hz with arrays separation 17 cm, and the microphone signal was recorded as a voltage on the oscilloscope at a distance of > 40 cm from the source.

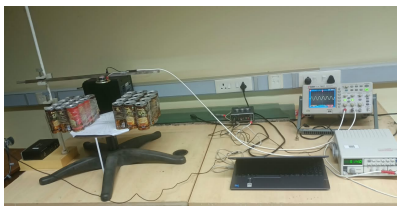
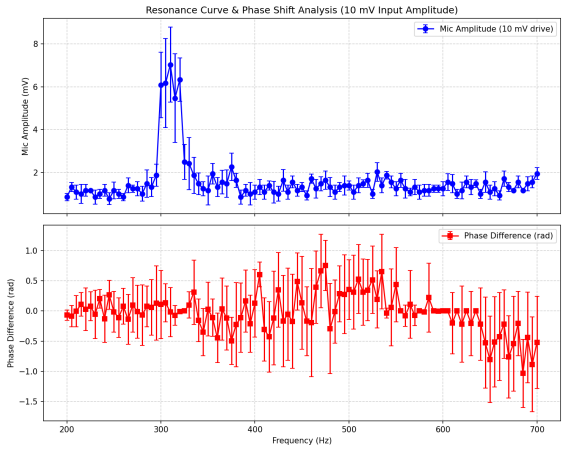


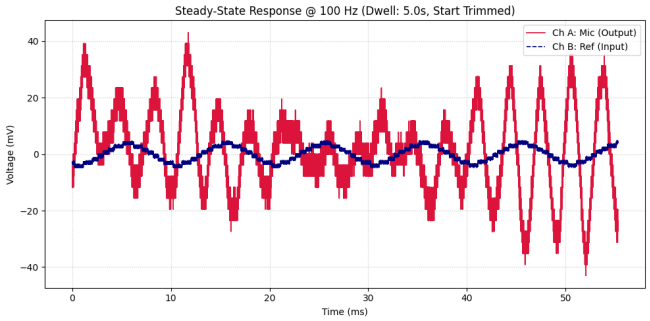
FIG. 4: Experimental setup of the acoustic meta-antenna array.

V. RESULT AND DATA ANALYSIS

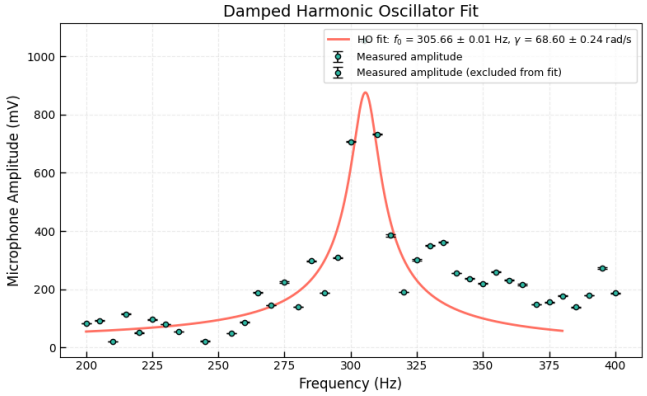
All Python codes used for data analysis via `picosdk`, signal processing, damped harmonic oscillator fitting, and figure generation were produced with assistance from AI tools.[3] The complete experimental data sheets, and generated plots are available at the GitHub repository:[4].



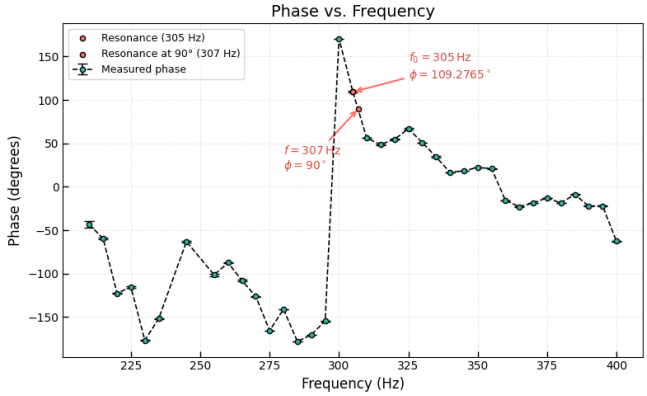
(a) Picoscope raw data.



(b) Data verification.



(c) Resonance curve.



(d) Phase difference versus frequency.

FIG. 5: Experimental Picoscope data and resonance analysis.

A. Part 1

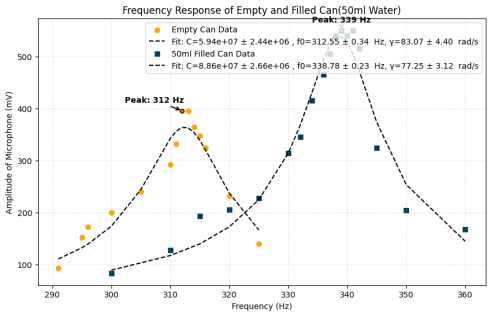
1. *Picoscope data analysis*

The resonance of the can (of 500ml volume) was first measured with a PicoScope using `picosdk` python package. The frequency was swept from 200 to 700 Hz with a 10 mV drive amplitude, and the microphone amplitude and its phase relative to the reference signal were recorded at each step. A resonance peak is visible near 305 Hz [Fig. 5a], but the error bars are large and the phase response is poor: the phase difference scatters widely about zero and does not show a clear transition through resonance.

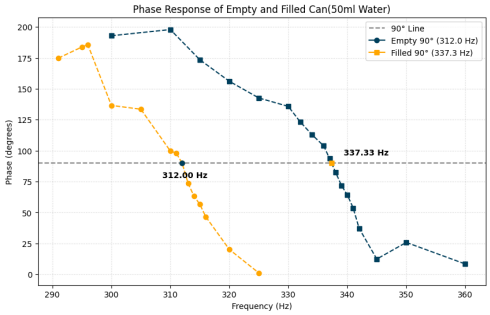
To find the origin of this noise, the input and output signals of the PicoScope were inspected directly in the time domain. Figure 5b shows the steady-state response at 100 Hz. The reference signal (channel B) is a clean sinusoid, whereas the microphone signal (channel A) is distorted and noisy, which indicates that the noise is added in the microphone channel, either by the amplifier or by the PicoScope input signal.

Since the noise level does not depend on the drive, we attempted to improve the signal-to-noise ratio by increasing the drive amplitude to 90 mV. This produced a considerably cleaner resonance curve and phase response, Figs. 5c and 5d. The amplitude-versus-frequency data were fitted using the single damped harmonic oscillator model given by Eq. (4), yielding a resonance frequency of $f_0 = 305.66 \pm 0.01$ Hz and a damping constant $\gamma_D = 68.60 \pm 0.24$ rad/s. Points far from resonance, where the signal is comparable to the background, were excluded from the fit (data included upto 380 Hz). The phase difference in Fig. 5d changes sharply at resonance, as expected for a driven oscillator.
However, noise remains visible in both the amplitude and the phase data. In addition, at the resonance frequency (305 Hz) the measured phase is 109°, instead of the expected 90°, which is reached only at 307 Hz. Because of this residual noise and phase offset, the measurement was repeated using an oscilloscope and a function generator, as described in the next section.

2. *Oscilloscope data analysis*



(a) Amplitude response.



(b) Phase response.

FIG. 6: Oscilloscope measurements for an empty can and a can filled with 50 ml of water. (a) Microphone amplitude with damped harmonic oscillator fits (dashed lines). (b) Phase difference versus frequency, with the 90° crossing marked for each can.

Using a function generator to drive the K5 behringer loudspeaker and an oscilloscope, the microphone and input signals were recorded. At each frequency, the amplitude of the microphone signal and its phase difference relative to the reference were observed from the oscilloscope. The step size was reduced to 1 Hz near resonance, which gives a denser sampling of the region where the amplitude and phase change most rapidly. The measurement was carried out for both an empty can (of volume 500 ml) and filled can containing 50 ml of water.

The amplitude responses [Fig. **??**] show a clear resonance peak for both cans, with a much cleaner line shape than in the PicoScope data. The maxima of the measured data lie at 312 Hz for the empty can and at 339 Hz for the filled can. The data were fitted using the damped harmonic oscillator model of Eq. (4), giving $f_0 = (312.55 \pm 0.34)$ Hz and $\gamma_D = (83.07 \pm 4.40)$ rad/s for the empty can, and $f_0 = (338.78 \pm 0.23)$ Hz and $\gamma_D = (77.25 \pm 3.12)$ rad/s for the filled can. Adding water reduces the volume of air in the can, so the resonance shifts to higher frequency by about 26 Hz, in agreement with the trend predicted by Eq. (2). The damping constants of the two cans are equal within about two standard errors, which is expected since the geometry of the neck is unchanged.

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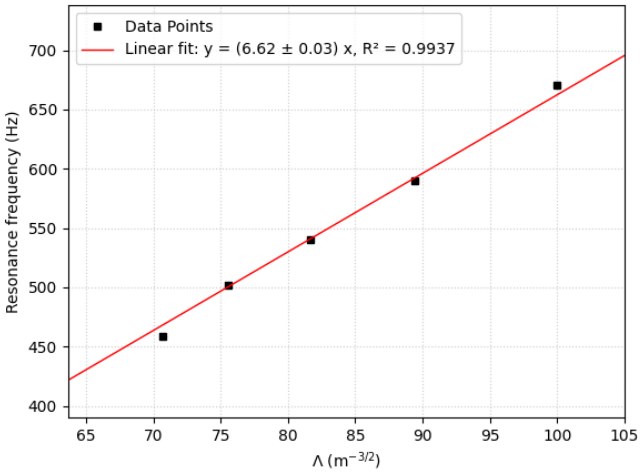
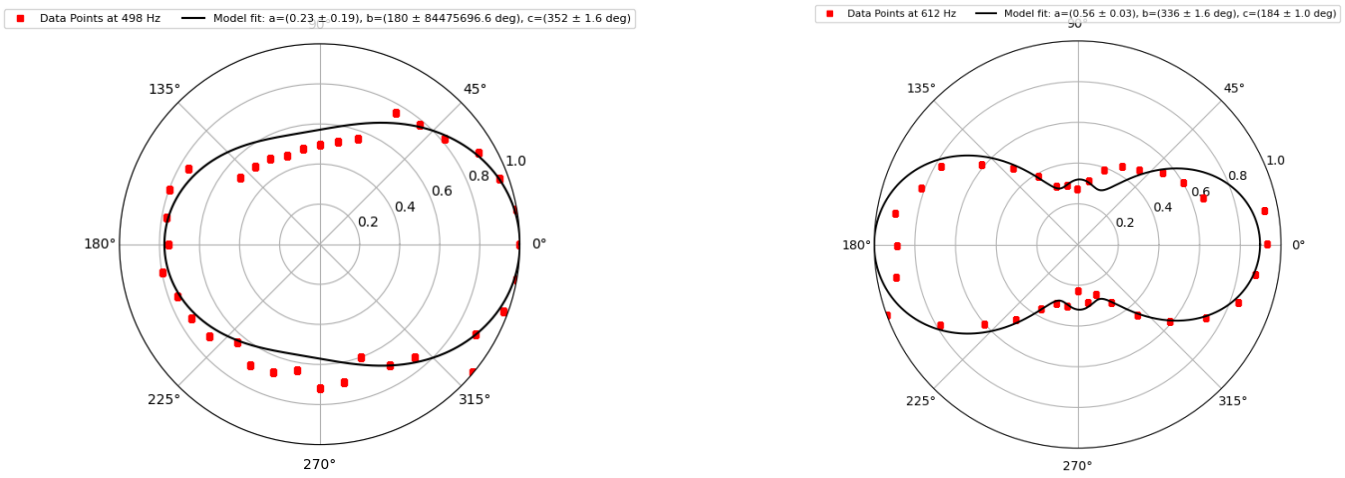


FIG. 7: Resonance frequency versus $\Lambda = 1/\sqrt{V_0}$ for different air volumes in the can. Squares are measured data; the line is a linear fit through the origin.

The phase difference [Fig. **??**] decreases through resonance by about 180° (i.e. π) for both cans but the behaviour of plot is not as expected from Eq.5. The frequency at which the phase equals 90° was found by linear interpolation between neighbouring points. For the empty can this occurs at 312.0 Hz, which coincides with the amplitude maximum. For the filled can it occurs at 337.3 Hz, about 1.5 Hz below the fitted resonance frequency of 338.8 Hz, which is a small offset compared with the 109° phase measured at resonance with the PicoScope. The residual scatter in the phase data is possibly due to the room noise and instrumental noise.
Also, to observe the resonance dependence on the internal volume, the can (from here 200 ml can was used) was progressively filled with 0-100 ml of water and the resonance frequency was measured at each step. The resonance frequency is plotted against $\Lambda = 1/\sqrt{V_0}$, where V_0 is the remaining air volume, the data follow a straight line through the origin [Fig. 7], as predicted by Eq. (2). A linear fit gives a slope of $6.62 \pm 0.03\,\mathrm{Hz\,m^{3/2}}$ with $R^2 = 0.994$. The small deviations at the lowest and highest volumes are expected, since the model assumes that the cavity volume is much larger than the neck volume.

B. Part 2

The polar distributions were measured with the two 4×3 arrays placed 17 cm apart. The microphone amplitude was recorded in 10° steps around the antenna and normalized to its maximum value. The data were fitted with the amplitude $|f(\theta)|$ of Eq. (10), written in terms of $a = A_1/A_0$, $b = (\phi_1 + \phi_2)/2$ and $c = (\phi_1 - \phi_2)/2$ [Fig. 8].



(a) 498 Hz. (b) 612 Hz.

FIG. 8: Polar distributions for arrays 17 cm apart at (a) 498 Hz and (b) 612 Hz. Red squares are measured data and black lines are fits of the three-source model.

The separation enters through $k\,\delta z$, with δz half the array separation. It depends on the driving frequency: at 498 Hz the 17 cm separation is $\approx \lambda/4$ ($k\,\delta z = \pi/4$), while at 612 Hz it is $\lambda/3.3$ ($k\,\delta z = \pi/3.3$).

At 498 Hz [Fig. 8(a)] the pattern is broad and only weakly directional: the signal is largest along 0° and the minimum near 110°. The fit reproduces this shape and gives a small array-to-loudspeaker amplitude ratio, $a = 0.35 \pm 0.10$, with $c = 172° \pm 1.7°$, i.e. a phase difference between the arrays of $\phi_1 - \phi_2 = 2c \approx 344° \equiv -16°$, close to zero. The two arrays therefore respond almost in phase with each other and weakly compared with the speaker.

At 612 Hz [Fig. 8(b)] the fit again follows the data and gives $a = 0.51 \pm 0.01$, $b = 172° \pm 0.5°$ and $c = 2° \pm 0.4°$. From the plot, it can be seen that sound waves are asymmetrically distributed. At 0° and 180°, the amplitude response is maximum while at 90° and 270° the amplitude is minimum as expected in the case of antenna. It shows that waves responding more in one direction than the other and so we can see the directionality behaviour of antenna.
In both cases, so a directional (cardioid-like) response was not obtained as phase different was not exactly π. The probable reason for the deviation could be that the discrepancies between the experimental data and the fitted curves could be due to reflections and standing waves in the room, size of the array compared to the wavelength, etc.

VI. CONCLUSION

In this experiment, we performed the wave manipulation characteristics of acoustic metamaterialsusing standard soda cans as subwavelength Helmholtz resonators. The characterizations performed on single cans proved the effectiveness of the damped driven harmonic oscillator model, showing the 90° phase shift due to resonance and confirming the inverse square root relationship between the frequency of resonance and the volume of air inside the can ($f_0 \propto 1/\sqrt{V_0}$ with $R^2 = 0.994$). Through arranging these meta-atoms into two 4×3 arrays, we were able to form a directional acoustic antenna and measure its angular radiation fields at 498 Hz and 612 Hz. However, because of environmental reflections, standing waves, and phase detuning, a perfect cardioid could not be formed.

VII. SOURCES OF ERROR

1. Non-Circular Aperture Geometry: the openings of cans are not perfectly circular.
2. Instrumental noise: either from picoscope, cables and oscilloscpe.
3. Manual water filling created slight resonance shifts.
4. Microphone placement near the can opening subtly altered localized airflow and effective acoustic neck mass.
5. External noise from the surroundings (from walls in the room).

VIII. APPLICATIONS

1. Directional Sound Beaming that enables targeted acoustic emission and directional sound projection without requiring large physical structures.

2. Resonator-based metasurfaces allow ultra-thin, lightweight panels engineered for efficient low-frequency sound absorption and noise cancellation.
3. Used as acoustic Lensing and Cloaking.

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[1] J. T. Wilkinson, C. B. Whitehouse, R. F. Oulton, and S. D. Gennaro, An undergraduate experiment demonstrating the physics of metamaterials with acoustic waves and soda cans, American Journal of Physics **84**, 14 (2016).

[2] Wikipedia contributors, Helmholtz resonance, `https://en.wikipedia.org/wiki/Helmholtz_resonance` (2026), accessed: 2026-10-05.

[3] Anthropic, Claude AI, `https://claude.ai/`, accessed: 12 October 2026.

[4] A. Arya, Open Lab NISER: Integrated lab, `https://github.com/Anuj-Arya1/Open-Lab-NISER/tree/main/INTEGRATED\%20LAB`, gitHub repository, accessed October 12, 2026.